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Segmented variable fan outlet guide vane with segment interface components

Abstract

A fan assembly for a gas turbine engine includes a fan duct, a fan, and an outlet guide vane assembly located in the fan duct axially downstream of the fan. The outlet guide vane assembly includes a first variable leading edge outlet guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a tip segment configured to rotate about a leading edge pitch axis and a hub segment located radially inward of and separate from the tip segment, the hub segment configured to independently rotate about the leading edge pitch axis relative to the tip segment. The assembly further includes an air manipulating member arranged radially between the tip and hub segments and at least one actuation assembly configured to rotate one of the tip and hub segments.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Application Ser. No. 63/452,876, filed on Mar. 17, 2023, the entire disclosure of which is incorporated herein by reference.

FIELD OF THE DISCLOSURE

(1) The present disclosure relates generally to gas turbine engines, and more specifically to fan assemblies of gas turbine engines.

BACKGROUND

(2) Gas turbine engines are used to power aircraft, watercraft, power generators, and the like. Gas turbine engines typically include an engine core having a compressor, a combustor, and a turbine. The compressor compresses air drawn into the engine and delivers high pressure air to the combustor. In the combustor, fuel is mixed with the high pressure air and is ignited. Products of the combustion reaction in the combustor are directed into the turbine where work is extracted to drive the compressor and, sometimes, an output shaft. Left-over products of the combustion are exhausted out of the turbine and may provide thrust in some applications.

(3) Gas turbine engines also typically include a fan positioned within an inlet duct of the gas turbine engine. The fan includes rotating blades that force air into the compressor section of the engine, as well as potentially providing additional thrust via forcing air around the engine core through bypass ducts. The fan blades may experience various operability issues due to factors such as variations in the intake airflow and pressure fluctuations within the inlet and the bypass ducts.

SUMMARY

(4) The present disclosure may comprise one or more of the following features and combinations thereof.

(5) A fan assembly for a gas turbine engine according to the present disclosure includes a fan duct arranged circumferentially around a central axis, a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades and reduce incidence between the outlet guide vane assembly and the fan exit air.

(6) The outlet guide vane assembly includes a first variable leading edge outlet guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, a first air manipulating member arranged radially between the first tip segment and the first hub segment and configured to manipulate at least a portion of the incoming fan exit air flowing over the leading edge portion, and at least one actuation assembly configured to selectively rotate the first tip segment about the leading edge pitch axis to a first pitch angle relative to the incoming fan exit air and configured to selectively rotate the first hub segment independent of the first tip segment about the leading edge pitch axis to a second pitch angle relative to the incoming fan exit air.

(7) In some embodiments, the at least one actuation assembly is configured to rotate the first tip

segment and the first hub segment to the first pitch angle and the second pitch angle which is different than the first pitch angle.

(8) In some embodiments, the at least one actuation assembly includes a first actuation assembly arranged radially outward of the leading edge portion and configured to rotate the first tip segment to the first pitch angle.

(9) In some embodiments, the first air manipulating member is arranged proximate to a radially inner side of the first tip segment and proximate to a radially outer side of the first hub segment.

(10) In some embodiments, a radial gap is formed between the radially inner side of the first tip segment and the radially outer side of the first hub segment, and wherein the first air manipulating member has a radial extent that is sized so as to seal off the incoming fan exit air from flowing through the radial gap.

(11) In some embodiments, the first air manipulating member is a winglet defining a radially outer surface that is generally planar and faces radially outwardly and a radially inner surface that is generally planar and faces radially inwardly, and wherein the winglet includes an axially aft end and an axially forward edge.

(12) In some embodiments, a width of the winglet is equal to or greater than a width of the first tip segment and the first hub segment along an entire axial extent of the first tip segment and the first hub segment.

(13) In some embodiments, a width of the winglet at the axially aft end is less than a length of the forward edge, the forward edge is curved so as to match a path of movement of a leading edge of at least one of the first tip segment and the first hub segment, the path of movement has a maximum first end point and a maximum second end point, and the length of the forward edge is equal to or extends beyond the maximum first end point and the maximum second end point of the path of movement of the at least one of the first tip segment and the first hub segment.

(14) In some embodiments, the axially aft end of the winglet is coupled to an axially forward end of the fixed aft portion.

(15) In some embodiments, the first air manipulating member is a seal, and wherein an outer perimeter of the seal generally corresponds to an outer perimeter of each of the first tip segment and the first hub segment.

(16) In some embodiments, the seal includes an axially aft end that is coupled to an axially forward end of the fixed aft portion.

(17) In some embodiments, the seal is attached to one of the radially inner side of the first tip segment and the radially outer side of the first hub segment.

(18) In some embodiments, the at least one actuation assembly further includes a second actuation assembly arranged radially inward of the leading edge portion and configured to rotate the first hub segment to the second pitch angle.

(19) In some embodiments, the first actuation assembly includes a first actuation head coupled to the first tip segment and a first actuation arm coupled to the first actuation head, rotation of the first actuation arm causes rotation of the first actuation head which causes rotation of first tip segment so as to rotate the first tip segment to the first pitch angle, the second actuation assembly includes a second actuation head coupled to the first hub segment and a second actuation arm coupled to the second actuation head, and rotation of the second actuation arm causes rotation of the second actuation head which causes rotation of first hub segment so as to rotate the first hub segment independent of the first tip segment to the second pitch angle.

(20) According to a further aspect of the present disclosure, a fan assembly for a gas turbine engine includes a fan duct arranged circumferentially around a central axis, a fan adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades and reduce incidence between the outlet guide vane assembly and the fan exit air.

- (21) The outlet guide vane assembly includes a first variable leading edge outlet guide vane including a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, a first air manipulating member arranged radially between the first tip segment and the first hub segment, and at least one actuation assembly configured to selectively rotate the first tip segment about the leading edge pitch axis to a first pitch angle relative to the incoming fan exit air and configured to selectively rotate the first hub segment independent of the first tip segment about the leading edge pitch axis to a second pitch angle relative to the incoming fan exit air.
- (22) In some embodiments, the at least one actuation assembly includes a first actuation assembly arranged radially outward of the leading edge portion and configured to rotate the first tip segment to the first pitch angle.
- (23) In some embodiments, the first air manipulating member is arranged proximate to a radially inner side of the first tip segment and proximate to a radially outer side of the first hub segment, a radial gap is formed between the radially inner side of the first tip segment and the radially outer side of the first hub segment, and the first air manipulating member has a radial extent that is sized so as to seal off the incoming fan exit air from flowing through the radial gap.
- (24) In some embodiments, the first air manipulating member is a winglet defining a radially outer surface that is generally planar and faces radially outwardly and a radially inner surface that is generally planar and faces radially inwardly, the winglet includes an axially aft end and an axially forward edge, a width of the winglet is equal to or greater than a width of the first tip segment and the first hub segment along an entire axial extent of the first tip segment and the first hub segment, and a width of the winglet at the axially aft end is less than a length of the forward edge.
- (25) In some embodiments, the first air manipulating member is a seal, and wherein an outer perimeter of the seal generally corresponds to an outer perimeter of each of the first tip segment and the first hub segment.
- (26) According to a further aspect of the present disclosure, a method comprises arranging a fan duct circumferentially around a central axis, providing a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, arranging an outlet guide vane assembly in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades and reduce incidence between the outlet guide vane assembly and the fan exit air, the outlet guide vane assembly including a first variable leading edge outlet guide vane extending radially relative to the central axis and including the leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, arranging a first air manipulating member radially between the first tip segment and the first hub segment and configured to manipulate at least a portion of the incoming fan exit air flowing over the leading edge portion, and arranging at least one actuation assembly relative to the first tip segment configured to selectively rotate the first tip segment about the leading edge pitch axis to a first pitch angle relative to the incoming fan exit air and configured to selectively rotate the first hub segment independent of the first tip segment about the leading edge pitch axis to a second pitch angle relative to the incoming fan exit air.
- (27) These and other features of the present disclosure will become more apparent from the following description of the illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a cutaway view of a gas turbine engine that includes a fan assembly having a fan having plurality of fan blades that extend radially outward relative to the central axis, an engine core having a compressor, a combustor, and a turbine, and an outlet guide vane assembly located in a fan duct axially downstream of the plurality of fan blades that is configured to reduce the incidence between the outlet guide vane assembly and the fan exit air received from the fan blades and return the flow to generally axial flow;
- (2) FIG. 2 is a side cross-sectional view of the gas turbine engine of FIG. 1, showing the fan assembly including the plurality of fan blades, showing that the engine further includes an outer casing and an inner wall that define a fan duct passage through which the fan exit air flows, showing that the outlet guide vane assembly includes a first variable leading edge outlet guide vane having a first tip segment and a first hub segment configured to more closely match the vane to distorted fan exit air, and showing that the first tip and hub segments are hinged to each other;
- (3) FIG. 3 is a side cross-sectional view of the first variable leading edge outlet guide vane of FIG. 2, showing that the first variable leading edge outlet guide vane includes the first tip segment and the first hub segment, and showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the leading edge portion and a second actuation assembly arranged radially inward of the leading edge portion;
- (4) FIG. 4A is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 3, showing the first hub segment rotated to a pitch angle and the first tip segment at a neutral position;
- (5) FIG. 4B is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 3, showing the first tip segment rotated to a pitch angle and the first hub segment at a neutral position;
- (6) FIG. 4C is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 3, showing the first hub segment and the first tip segment rotated to differing pitch angles;
- (7) FIG. 5A is a side cutaway perspective view of a plurality of the first variable leading edge outlet guide vanes of FIG. 3, showing that the first actuation heads are connected to a full annular ring via actuation arms;
- (8) FIG. 5B is a side cutaway perspective view of a plurality of the first variable leading edge outlet guide vanes of FIG. 3, showing that two groups of first actuation heads are connected to annular ring segments via actuation arms, the segments being spaced apart circumferentially;
- (9) FIG. 6 is a side cutaway perspective view of a plurality of the first variable leading edge outlet guide vanes of FIG. 3, showing that the first actuation heads are each connected to individual actuators, which are connected to a control system;
- (10) FIG. 7 is a side cross-sectional view of the first variable leading edge outlet guide vane of FIG. 3, showing that the first tip and hub segments are not hinged to each other;
- (11) FIG. 8 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes a first tip segment and a first hub segment, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the leading edge portion and a second actuation assembly arranged radially inward of the leading edge portion, and showing that the guide vane further includes a first air manipulating member arranged radially between the first tip segment and the first hub segment;
- (12) FIG. 9 is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 8, showing the first tip segment at a neutral position and a top view of the first air manipulating member configured as a winglet;
- (13) FIG. 10 is a perspective view of the first variable leading edge outlet guide vane of FIG. 8, showing the first air manipulating member configured as a winglet;

(14) FIG. 11 is a front perspective view of the first variable leading edge outlet guide vane of FIG. 8, showing the first air manipulating member configured as a winglet;

(15) FIG. 12 is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 8, showing a top view of the first air manipulating member configured as a seal;

(16) FIG. 13 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes a first tip segment and a first hub segment, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the leading edge portion and a second actuation assembly arranged radially inward of the leading edge portion, and showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment;

(17) FIG. 14 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes a first tip segment and a first hub segment, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the leading edge portion and a second actuation assembly arranged radially inward of the leading edge portion, showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment, and showing that the guide vane further includes air manipulating members arranged radially between the first tip segment and the central portion and between the first hub segment and the central portion;

(18) FIG. 15 is a perspective view of the first variable leading edge outlet guide vane of FIG. 14, showing the air manipulating members configured as a winglets;

(19) FIG. 16A is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes a first tip segment and a first hub segment, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the leading edge portion, and showing that the first actuation assembly includes a first actuation head coupled to the first tip segment so as to rotate the first tip segment, a second actuation head radially outward of the first actuation head, and a first actuation rod extending through the first actuation head from the second actuation head to the first hub segment so as to rotate the first hub segment;

(20) FIG. 16B is a side cross-sectional view of a first variable leading edge outlet guide vane of FIG. 16A, showing an alternative actuation arrangement of the second actuation head;

(21) FIG. 16C is a side cross-sectional view of a first variable leading edge outlet guide vane of FIG. 16A, showing an alternative actuation arrangement of the second actuation head;

(22) FIG. 17 is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 16A, showing the first tip segment at a neutral position and the first actuation head and the actuation rod;

(23) FIG. 18 is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 16A, showing the first tip segment at a neutral position and the first hub segment rotated to a pitch angle;

(24) FIG. 19 is a top cross-sectional view of the first variable leading edge outlet guide vane of FIG. 16A, showing the first hub segment at a neutral position and the first tip segment rotated to a pitch angle;

(25) FIG. 20 is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes a first tip segment and a first hub segment, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the leading edge portion, showing that the first actuation assembly includes a first actuation head coupled to the first tip segment so as to rotate the first tip segment, a second actuation head radially outward of the first actuation head, and a first actuation rod extending through the first actuation head from the second

actuation head to the first hub segment so as to rotate the first hub segment, and showing that the guide vane further includes a first air manipulating member arranged radially between the first tip segment and the first hub segment;

(26) FIG. **21** is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes a first tip segment and a first hub segment, showing that the outlet guide vane assembly includes a first actuation assembly arranged radially outward of the leading edge portion, showing that the first actuation assembly includes a first actuation head coupled to the first tip segment so as to rotate the first tip segment, a second actuation head radially outward of the first actuation head, and a first actuation rod extending through the first actuation head from the second actuation head to the first hub segment so as to rotate the first hub segment, and showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment;

(27) FIG. **22** is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes first and second tip segments and first and second hub segments, showing that the outlet guide vane assembly includes a first actuation assembly including a first actuation head coupled to the first tip segment, a second actuation head, and a first actuation rod extending through the first actuation head from the second actuation head to the second tip segment so as to rotate the second tip segment, showing that the outlet guide vane assembly includes a second actuation assembly including a third actuation head coupled to the first hub segment, a fourth actuation head, and a second actuation rod extending through the third actuation head from the fourth actuation head to the second hub segment so as to rotate the second hub segment, and showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment;

(28) FIG. **23** is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes first and second tip segments and first and second hub segments, showing that the outlet guide vane assembly includes a first actuation assembly including a first actuation head coupled to the first tip segment, a second actuation head, and a first actuation rod extending through the first actuation head from the second actuation head to the second tip segment so as to rotate the second tip segment, showing that the outlet guide vane assembly includes a second actuation assembly including a third actuation head coupled to the first hub segment, a fourth actuation head, and a second actuation rod extending through the third actuation head from the fourth actuation head to the second hub segment so as to rotate the second hub segment, showing that the guide vane further includes a static central portion arranged radially between the first tip segment and the first hub segment, and showing that air manipulating members arranged radially between the tip and hub segments;

(29) FIG. **24** is a perspective view of the first variable leading edge outlet guide vane of FIG. **23**, showing the air manipulating members configured as a winglets;

(30) FIG. **25** is a side cross-sectional view of a first variable leading edge outlet guide vane according to a further aspect of the present disclosure, showing that the first variable leading edge outlet guide vane includes first and second tip segments and first and second hub segments, showing that the outlet guide vane assembly includes a first actuation assembly including a first actuation head coupled to the first tip segment, a second actuation head, and a first actuation rod extending through the first actuation head from the second actuation head to the second tip segment so as to rotate the second tip segment, showing that the outlet guide vane assembly includes a second actuation assembly including a third actuation head coupled to the first hub segment, a fourth actuation head, and a second actuation rod extending through the third actuation head from the fourth actuation head to the second hub segment so as to rotate the second hub segment, and

showing that the guide vane does not include a static central portion arranged radially between the first tip segment and the first hub segment;

(31) FIG. **26** is a graph showing distribution of span percentage versus inlet segment angle for a generalized matched vane versus a nominal vane according to the present disclosure;

(32) FIG. **27** is a graph showing span percentage versus inlet segment angle for rotation of a split nominal vane to resemble variant distributions for a variant vane assembly according to the present disclosure, including four segments and a static central portion;

(33) FIG. **28** is a graph showing span percentage versus inlet segment angle for a nominal vane and rotated segments of the nominal vane for a four segment vane assembly with a static central portion according to the present disclosure;

(34) FIG. **29** is a graph showing span percentage versus inlet segment angle for a generalized section variation for a two segment vane assembly having a static central portion according to the present disclosure; and

(35) FIG. **30** is a graph showing span percentage versus inlet segment angle for a generalized section variation for a four segment vane assembly having a static central portion according to the present disclosure.

DETAILED DESCRIPTION

(36) For the purposes of promoting an understanding of the principles of the disclosure, reference will now be made to a number of illustrative embodiments illustrated in the drawings and specific language will be used to describe the same.

(37) An illustrative aerospace gas turbine engine **10** includes a fan assembly **12** and an engine core **13** having a compressor **14**, a combustor **16**, and a turbine **18**, as shown in FIG. **1**. The fan assembly **12** is driven by the turbine **18** and provides thrust for propelling an air vehicle by forcing fan exit air **15** through a fan duct **20** that circumferentially surrounds an outer casing **17** of the engine core **13**. The compressor **14** compresses and delivers air to the combustor **16**. The combustor **16** mixes fuel with the compressed air received from the compressor **14** and ignites the fuel. The hot, high-pressure products of the combustion reaction in the combustor **16** are directed into the turbine **18** to cause the turbine **18** to rotate about a central axis **11** and drive the compressor **14** and the fan **12**. In some embodiments, the fan may be replaced with a propeller, drive shaft, or other suitable configuration.

(38) The fan assembly **12** includes a fan **21** having a plurality of fan blades **22** that extend radially outward relative to the central axis **11** and that are located in the inlet of the gas turbine engine **10**, as shown in FIGS. **1** and **2**. The fan blades **22** direct at least a portion of the air flowing over the blades **22**, this portion being fan exit air **15** as shown in FIGS. **1** and **2**, through the fan duct **20** such that the fan exit air **15** bypasses the engine core **13** and provides additional thrust for the gas turbine engine **10**. The fan duct **20** includes an outer fan duct casing **19** and an inner wall **23** that together define an annular fan duct passage **24** through which the fan exit air **15** flows and subsequently exits the fan duct **20** into the ambient air surrounding the engine **10**. The inner wall **23** may include an axially forward end **23S** that functions as a splitter such that a portion of the incoming fan exit air **15** enters the engine core through the engine core passage **23P** radially inward of the inner wall **23** and a portion of the fan exit air **15** enters the fan duct **20**.

(39) In the illustrative embodiment, the fan assembly **12** further includes outlet guide vane assembly **28** located in the fan duct **20** axially downstream of the plurality of fan blades **22** that is configured to reduce the incidence between the outlet guide vane assembly **28** and the fan exit air **15** received from the plurality of fan blades **22** and return the flow to generally axial flow, as shown in FIG. **2**. In some embodiments, the outlet guide vane assembly **28** is arranged axially downstream of the axially forward end **23S** of the inner wall **23**. In the illustrative embodiment, the outlet guide vane assembly **28** includes a first variable leading edge outlet guide vane **30** having a leading edge portion **31** and a fixed aft portion **50**, as shown in FIG. **3**. The leading edge portion **31** includes a first tip segment **32** configured to rotate about a leading edge pitch axis **39** and a first hub segment

42 located radially inward of and separate from the first tip segment **32**, also configured to rotate about the leading edge pitch axis **39**.

(40) The first variable leading edge outlet guide vane **30** extends radially outward relative to the central axis **11**, as shown in FIG. 3. In some embodiments, the first variable leading edge outlet guide vanes **30** may include a plurality of first variable leading edge outlet guide vanes **30** disposed around a circumferential extent of the inner wall **23** define a first outlet guide vane stage.

(41) As shown in FIG. 3 and contextualized in FIG. 4A, the first variable leading edge outlet guide vane **30** includes an airfoil shape. The first tip segment **32** includes a leading edge **33** located at a forward end of the segment **32**, a trailing edge **34** axially spaced apart from the leading edge **33** and located at an aft end of the segment **32**, a radially outer side **35**, and a radially inner side **36**. Similarly, the first hub segment **42** includes a leading edge **43** located at a forward end of the segment **42**, a trailing edge **44** axially spaced apart from the leading edge **43** and located at an aft end of the segment **42**, a radially outer side **45**, and a radially inner side **46**.

(42) Together, the first tip and hub segments **32**, **42** form the leading edge portion **31**. As shown in FIG. 4A, the leading edge portion **31** and the fixed aft portion **50** together form the overall vane **30**, which includes a pressure side surface **31P** that extends between the leading edges **33**, **43** of the segments **32**, **42** and a trailing edge **52** of the fixed aft portion **50** on one side of the vane **30**.

Similarly, a suction side surface **31S** extends between the leading edges **33**, **43** of the segments **32**, **42** and a trailing edge **52** of the fixed aft portion **50** on an opposing side of the vane **30**.

Illustratively, the leading edge portion **31** is approximately one-half of the total chord length **30H** of the vane **30**, while the fixed aft portion **50** is approximately one-half of the total chord length **30H**, as shown in FIG. 4A. In other embodiments, the leading edge portion **31** is approximately one-third of the total chord length **30H** of the vane **30**, while the fixed aft portion **50** is approximately two-thirds of the total chord length **30H**.

(43) The first tip and hub segments **32**, **42** are configured to rotate about the leading edge pitch axis **39**, as shown in FIG. 3. In the illustrative embodiment, the leading edge pitch axis **39** is located closer to the trailing edges **34**, **44** of the segments **32**, **42** than the leading edges **33**, **43**. In this manner, the first tip and hub segments **32**, **42** can rotate proximate to the fixed aft portion **50** such that airflow remains uninterrupted while flowing over the vane **30**.

(44) The first hub segment **42** is configured to independently rotate about the leading edge pitch axis **39** relative to the first tip segment **32**, and may include a small radial gap therebetween. In some embodiments, the first hub segment **42** can include a hinge rod **47** extending radially outwardly from the radially outer side **45** of the first hub segment **42**. The first tip segment **32** can include a hinge rod receiving cavity **37** configured to securely receive the hinge rod **47** so as to add stability to the rotation of the two segments **32**, **42** relative to each other while allowing for independent rotation. The hinge rod **47** of the first hub segment **42** and the hinge rod receiving cavity **37** of the first tip segment **32** can each be cylindrical and aligned with the leading edge pitch axis **39**. In some embodiments in which the first tip and hub segments **32**, **42** are formed of a sufficiently stiff material, a hinge rod may be omitted, as shown in FIG. 7.

(45) To rotate independently, the illustrative embodiment of the outlet guide vane assembly **28** includes a first actuation assembly **70** associated with the first tip segment **32** and a second actuation assembly **80** associated with the first hub segment **42**. The first actuation assembly **70** is arranged radially outward of the leading edge portion **31** and includes a first actuation arm **74** and a first actuation head **76**, the first actuation arm **74** extending axially away from the first actuation head **76**. The second actuation assembly **80** is arranged radially inward of the leading edge portion **31** and includes a second actuation arm **84** and a second actuation head **86**, the second actuation arm **84** extending axially away from the second actuation head **86**.

(46) The first actuation head **76** extends through the outer fan duct casing **19** and is fixedly coupled to the radially outer side **35** of the first tip segment **32**, as shown in FIG. 3. Specifically, the first actuation head **76** is aligned with the leading edge pitch axis **39** such that rotation of the first

actuation head **76** rotates the first tip segment **32** about the leading edge pitch axis **39**. In some embodiments, the actuation head **76** is cylindrical having a central axis that aligns with the leading edge pitch axis **39**. The first actuation head **76** is fixedly arranged within an opening **77** formed at a first end **74A** of the first actuation arm **74**. As a result, pivoting movement of the first actuation arm **74** about the leading edge pitch axis **39** rotates the first actuation head **76**, which in turn rotates the first tip segment **32**. In some embodiments, the first actuation arm **74** is arranged radially outward of the vane **30** and the outer fan duct casing **19**, as shown in FIG. 3.

(47) The second actuation head **86** extends through the inner wall **23** and is fixedly coupled to the radially inner side **46** of the first hub segment **42**, as shown in FIG. 3. Specifically, the second actuation head **86** is aligned with the leading edge pitch axis **39** such that rotation of the second actuation head **86** rotates the first hub segment **42** about the leading edge pitch axis **39**. In some embodiments, the actuation head **86** is cylindrical having a central axis that aligns with the leading edge pitch axis **39**. The second actuation head **86** is fixedly arranged within an opening **87** formed at a first end **84A** of the second actuation arm **84**. As a result, pivoting movement of the second actuation arm **84** about the leading edge pitch axis **39** rotates the second actuation head **86**, which in turn rotates the first hub segment **42**. In some embodiments, the second actuation arm **84** is arranged radially inward of the vane **30** and the inner wall **23**, as shown in FIG. 3. The inner wall **23** may not define a radially outer wall or boundary of the engine core passage **23P** such that the components of the second actuation assembly **80** can be arranged radially outward of the engine core passage **23P** and not interrupt airflow therethrough.

(48) As will be described in greater detail below, the first and second actuation arms **74**, **84** are configured to be moved generally circumferentially relative to the first and second actuation heads **76**, **86**, respectively, by annular rings, segmented annular rings, or individual actuators so as to pivot the actuation arms **74**, **84** about the leading edge pitch axis **39**. In particular, the actuation arms **74**, **84** may be selectively pivoted about the leading edge pitch axis **39** so as to rotate the respective actuation head **76**, **86** so as to selectively rotate the first tip and hub segments **32**, **42** about the leading edge pitch axis **39**. As a result, the first tip segment **32** can be selectively rotated about the leading edge pitch axis **39** to a first pitch angle relative to the incoming fan exit air **15**. Similarly, the first hub segment **42** can be selectively rotated about the leading edge pitch axis **39** to a second pitch angle relative to the incoming fan exit air **15**. The fixed trailing edge **50** rotates any flow back to near axial to minimize loss from swirl.

(49) In this way, the first tip and hub segments **32**, **42** can be individually controlled with respect to each other to the same or differing angles, thus allowing for a multitude of segment positions and arrangements to be achieved. This provides for great flexibility in managing incoming airflows which may include distortions and disturbances. This can be particularly useful in embedded engine applications with complex intake and inlet duct geometries. In such scenarios, the distortion flows have more significant gradients and vortices, even in flight, so accommodation is necessary to maintain fan operability and performance.

(50) In one exemplary arrangement, as shown in FIG. 4A, the first hub segment **42** can be rotated to a first pitch angle and the first tip segment **32** may remain at a neutral position. In another exemplary arrangement, as shown in FIG. 4B, the first tip segment **32** can be rotated to a first pitch angle and the first hub segment **42** may remain at a neutral position. In another exemplary arrangement, as shown in FIG. 4C, the first tip and hub segments **32**, **42** can both be rotated to different pitch angles.

(51) As shown in FIG. 3, and in greater detail in FIGS. 5A, 5B, and 6, the outlet guide vane assembly **28** can further include a plurality of variable leading edge outlet guide vanes **30** arranged around the entirety of the circumferential extent of the vane assembly **28**. The plurality of variable leading edge outlet guide vanes **30** include the first variable leading edge outlet guide vane **30** described above. Each variable leading edge outlet guide vane **30** of the plurality of variable leading edge outlet guide vanes **30** includes a respective tip segment **32** and hub segment **42**, as

well as respective actuation assemblies **70**, **80**. The first and second actuation arms **74**, **84** of each vane **30** may be coupled to respective annular rings (for example, first and second annular rings **62**, **64**) that extend circumferentially about the central axis **11**. The actuation arms **74**, **84** may be coupled to the rings **62**, **64** via an opening formed at a second end **74B**, **84B** of each actuation arm **74**, **84**.

(52) As can be seen in FIG. 5A, the outlet guide vane assembly **28** may include a fully annular first annular ring **62** and a fully annular second annular ring **64**. In this configuration, the first and second annular rings **62**, **64** can be configured to rotate about the central axis **11** so as to move each actuation arm **74**, **84** coupled thereto generally circumferentially relative to the first and second actuation heads **76**, **86**, or in other words, to pivot each actuation arm **74**, **84** coupled thereto about the leading edge pitch axis **39**. As a result, the rotation of the first and second annular rings **62**, **64** about the central axis **11** causes rotation of the first tip and hub segments **32**, **42**, respectively. The first annular ring **62** can be rotated to a different rotational position as compared to the second annular ring **64** so as to rotate the first tip segment **32** to a different pitch angle than the first hub segment **42**. As will be described in detail below, the annular rings **62**, **64** may be controlled by a control system **90**.

(53) In some scenarios, it may be beneficial to have more control over individual sections of the plurality of variable leading edge outlet guide vanes **30** or individual vanes **30**. This may be particular useful when the engine experiences more significantly distorted flows, such as in embedded engine applications. Sectional control or individual control would provide additional flexibility to accommodate as much flow variation as possible to recover operability and performance margins. For example, as an aircraft maneuvers through ground crosswind or variations of flight orientations such as sideslip and pitch variations, the plurality of variable leading edge outlet guide vanes **30** can be rotationally moved in sections or independently to best reset the inlet angles and maximize the fan operability envelope. The optimized rest of the inlet angles improves stall margin, improves efficiency and performance, and reduces fan forcing debits.

(54) As can be seen in FIG. 5B, first annular ring **62** may be broken up into segments so as to include multiple circumferential segments disposed about the central axis **11**. In particular, the multiple circumferential segments are radially aligned with each other and circumferentially spaced apart. For example, the assembly **28** may include a first annular ring segment **62S1** and a second annular ring segment **62S2** radially aligned with the first annular ring segment **62S1**, extending partially circumferentially about the central axis **11**, and circumferentially offset from the first annular ring segment **62S1**. Similarly, the assembly **28** may include a third annular ring segment **64S1** and a second annular ring segment **64S2** (shown in phantom) radially aligned with the third annular ring segment **64S1**, extending partially circumferentially about the central axis **11**, and circumferentially offset from the third annular ring segment **64S1**.

(55) In this configuration, the first and second annular ring segments **62S1**, **62S2** can be configured to independently rotate about the central axis **11** so as to move each actuation arm **74**, **84** coupled thereto generally circumferentially relative to the first and second actuation heads **76**, **86**, or in other words, to pivot each actuation arm **74**, **84** coupled thereto about the leading edge pitch axis **39**. The first and second annular ring segments **62S1**, **62S2** are circumferentially spaced apart a great enough distance to allow maximum movement relative to each other. For example, in some scenarios, the first annular ring segment **62S1** may be moved in an opposing circumferential direction to the second annular ring segment **62S2** such that the respective tip segments **32** of their associated vanes **30** are rotated in opposing rotational directions. The third and fourth annular ring segments **64S1**, **64S2** can be similarly configured.

(56) The outlet guide vane assembly **28** can include multiple annular ring segments similar to the segments **62S1**, **62S2**, **64S1**, **64S2** disposed around the entire circumferential extent of the assembly **28**. As such, the multiple annular ring segments can define groups of vanes **30**, the tip and hub segments **32**, **42** of each being able to be rotated to unique pitch angles based on the

circumferential movement and position of their respective annular ring segment. As will be described in detail below, the annular rings **62**, **64** may be controlled by a control system **90**.

(57) In some embodiments, the control system **90** is configured to rotate each segment **32**, **42** of each vane **30** of the first plurality of variable leading edge outlet guide vanes **30** individually relative to each other vane's **30** segments **32**, **42**, as shown in FIG. **6**. That is to say, each segment **32**, **42** of each vane **30** may be rotated without moving any of the other vanes' segments **32**, **42** of the first and second plurality of variable leading edge outlet guide vanes **30**. This allows for the vanes **30** to be controlled in a variety of configurations. For example, a group of the plurality of variable leading edge outlet guide vanes **30** may be controlled to be rotated in unison and rotated to a first pitch angle, while other vanes **30** are individually controlled, each to unique pitch angles. The individual controllability of the vanes **30** accounts for variations in the fan exit air **15** around the circumference of the area of the plurality of variable leading edge outlet guide vanes **30**, which is particularly beneficial for the reasons discussed above.

(58) In another example, if the rotation of the first plurality of guide vanes **30** causes more undesirable flow effects in certain circumferential sectors, only vanes **30** located in those certain circumferential sectors may be rotated to specific pitch angles to reduce losses from said flow effects, while other vanes **30** may be rotated to different pitch angles. Then, if the flow dynamically encounters different distortions during operation, different circumferential sectors may require vane **30** adjustment. The individual controllability allows for the option to adjust these different circumferential sectors dynamically.

(59) In order to carry out the individual control, the second ends **74B**, **84B** of the actuation arms **74**, **84** may include a control member **75** coupled to an external actuator (not shown) that moves the actuation arms **74**, **84** generally circumferentially relative to the first and second actuation heads **76**, **86**. In other embodiments, the first and second actuation heads **76**, **86** may be individually actuated via an actuator coupled directly to the actuation head **76**, **86**, for example an actuator coupled to a radially outer side of the first actuation head **76** and an actuator coupled to a radially inner side of the second actuation head **86**.

(60) As touched on above, by controlling the segments **32**, **42** of at least one vane **30** of the plurality of variable leading edge outlet guide vanes **30**, the control system **90** is configured to control at least some of the flow of the fan exit air **15** after it passes over and exits the fan blades **22**. By controlling the entirety of the plurality of variable leading edge outlet guide vanes **30**, the control system **90** can accommodate the overall flow of the fan exit air **15**, in particular distorted flow, after it passes over and exits the fan blades **22** in order to control fan blade **22** response to forces acting on the fan blades **22**, as well as to reduce losses created by undesirable variations in the air flow. Moreover, because the fan exit air **15** may not be uniform as it exits the fan blades **22**, the plurality of variable leading edge outlet guide vanes **30** or the axial passage between the vanes **30** and the fan blades **22** operate further from their ideal design conditions. By adjusting the plurality of variable leading edge outlet guide vanes **30**, parameters such as incidence are improved, and detrimental flow conditions and losses in the vanes **30** or the axial passage between the vanes **30** and the fan blades **22** such as fan forcing and stall margin losses are reduced.

(61) In some embodiments, the control system **90** is configured to rotate the segments **32**, **42** of each vane **30** of the first plurality of variable leading edge outlet guide vanes **30** to a first vane-pitch angle in response to the gas turbine engine **10** operating at a given operating condition so as to reduce the incidence between the outlet guide vane assembly **28** and the fan exit air **15** received from the plurality of fan blades **22** and redirect the fan exit air **15** in a first direction, in particular a generally axial direction. In particular, the operating condition in which the fan assembly **12** and gas turbine engine **10** are operating in may include at least one of take-off, climb, cruise, descent, landing, and aircraft maneuvers of an aircraft having the engine **10** equipped. In each of these operating conditions, the plurality of fan blades **22** and/or the vanes **30** of the fan assembly **12** may experience various undesirable operability issues such as forcing, stall, and flutter. For example, the

engine **10** may operate in particular speed ranges for each of the operating conditions, and as result, the fan blades **22** may experience greater or lower levels of forcing, stall, and/or flutter in response to the engine **10** operating in particular speed ranges.

(62) In order to compensate for these forces acting on the fan blades **22**, the control system **90** is configured to rotate the segments **32**, **42** of the first plurality of variable leading edge outlet guide vanes **30** to an arrangement of first vane-pitch angles in order to alter the angle of the flow of fan exit air **15** after it exits the fan blades **22**. This change in the angle of flow as the fan exit air **15** passes over the first plurality of variable leading edge outlet guide vanes **30** reduces the amount of forcing, stall, and/or flutter experienced by the fan blades **22** and/or the outlet guide vanes **30**. Moreover, the control system **90** is configured to reset a desired incidence of air flow into the first plurality of variable leading edge outlet guide vanes **30** in response to swirl in the inlet flow. This produces an averaging effect that improves engine performance and efficiency. These arrangements of the vanes **30** can also recover the losses created by flow separation, flow distortions, vortices, and/or swirl.

(63) The control system **90** is operable to control the segments **32**, **42** of the first plurality of variable leading edge outlet guide vanes **30** in a variety of configurations and arrangements in order to compensate for inlet pressure distortion, vortices and swirl, thus reducing the forcing, stall, flutter, flow separation, and any other undesirable effects in the fan rotor or outlet vanes. For example, in some embodiments, the control system **90** is configured to rotate each vane **30** of the first plurality of guide vanes **30** in unison. In other words, all of the first plurality of guide vanes **30** move to the same first vane-pitch angle. In such embodiments, the each vane **30** may be mechanically connected to each other via the first and second annular rings **62**, **64**.

(64) In some embodiments, the control system **90** is configured to rotate at least two different groups of the segments **32**, **42** of variable leading edge outlet guide vanes **30**. For example, the control system **90** may be configured to selectively rotate each group of vanes **30** to create non-uniform backpressure that drives the fan inlet distortion flows within the fan to change or redistribute around the circumference of the fan. This locally reduces loading on fan blades **22** within a lip separated flow with low local pressure to reduce forcing and/or improve the uniformity of flow in general through the fan to reduce forcing. In particular, fully opening (allowing full flow through the guide vanes) at least one group of vanes **30** and fully closing at least one further group of vanes **30** (allowing no flow through the guide vanes) reduces a tendency for a local stall of the fan blades **22** that could lead to early overall stall in the fan. In some embodiments, the control system **90** is configured to rotate a large group of vanes **30** which counters bulk swirling flows or local changes to improve localized intake swirl gradients to improve fan performance and operability.

(65) In at least one additional embodiment, the plurality of variable leading edge outlet guide vane **30** are broken into unique groups of vanes **30**, as shown in FIG. 5B. Each group of vanes **30** is mechanically connected to each other, or ganged, via a unique circumferentially extending annular ring segment, such as the annular ring segments **62S1**, **62S2**, **64S1**, **64S2** shown in FIG. 5B. Although the illustrative embodiment shows each group of vanes **30** including two vanes **30**, the vanes **30** may be grouped and ganged in any combination of at least two groups of vanes totaling at least one fewer vane than the total number of vanes **30** in the plurality of outlet guide vanes **30**. For example, if the first plurality of outlet guide vanes **30** includes 60 vanes, a first group may include 30 vanes and a second group may include 30 vanes. As a further non-limiting example, a first group of vanes may include 50 vanes, a second group of vanes may include five vanes, and a third group of vanes may include five vanes.

(66) In some embodiments, the first plurality of variable leading edge outlet guide vanes **30** includes a first group of first vanes **30** and a second group of first vanes **30** different from the first group of guide vanes **30**. The control system **90** is configured to rotate the first group of first vanes **30** to a first vane-pitch angle and the second group of first vanes **30** to a second vane-pitch angle

that is different from the first vane-pitch angle. The groups of vanes **30** may be individually controlled or each group may be ganged together. For example, in some embodiments, one half of the first plurality of outlet guide vanes **30** is the first group and the other half of the first plurality of outlet guide vanes **30** is the second group.

(67) In some embodiments, the control system **90** utilizes predetermined arrangements of the plurality of variable leading edge outlet guide vanes **30** that are based on predetermined measurements and data taken in predetermined engine operating conditions and predetermined airflow characteristics. As such, the control system **90** is configured to rotate the vanes **30** to specific predetermined arrangements based on the operating condition and/or airflow characteristic(s) of the fan exit air **15** or the inlet air that the engine **10** and fan assembly **12** are operating in, or based on projected operating conditions and/or airflow characteristic(s) that will be encountered by the engine **10** during a mission.

(68) In some embodiments, the predetermined arrangements of the segments **32**, **42** of the plurality of variable leading edge outlet guide vanes **30** can be based on previously acquired test data corresponding to specific flight conditions. In a more complex arrangement, the control system **90** could be coupled to measurement systems, such as the at least one sensor **92** described below, that detect flow instabilities associated with impending fan stall to direct vane geometry changes to extend margins. The control system **90** may be more effective near stall as the vanes **30** can redistribute flow conditions to minimize local stall cells. Having smaller ganged vane **30** groups are efficient as well in that such groups only reduce losses locally to extend operability and performance.

(69) In some embodiments, the control system **90** includes at least one sensor **92** configured to take real-time measurements of the air flow within the fan duct passage **24** and of forces acting on the fan assembly components, as shown in FIG. 2. The real-time measurements may be utilized in order to determine the operating condition and/or airflow characteristic(s) of the fan exit air **15** or the inlet air that the engine **10** and fan assembly **12** are operating in so as to inform the control system **90** to which predetermined arrangement to rotate the plurality of variable leading edge outlet guide vanes **30**.

(70) In some embodiments, the control system **90** includes a neural network configured to perform machine learning such that the control system **90** can iterate over the predetermined arrangements in order to calculate new arrangements that are applicable to new variations in the operating condition and/or airflow characteristics that are unaccounted for by the predetermined settings and arrangements.

(71) In some embodiments, the control system **90** further includes a subsystem control that is integrated with other engine controls to further control reduction of losses created by undesirable variations in the air flow and improve engine performance and efficiency. For example, if rotation of vanes **30** resulted in a fan flow drop, the subsystem control is configured to compensate for this by increasing the fan speed in order to maintain thrust, and/or by changing the exhaust area of the engine **10** in order to further reduce the losses and improve engine efficiency.

(72) In some embodiments, the at least one sensor **92** may be located proximate to the fan blades **22**, proximate to the plurality of variable leading edge outlet guide vanes **30**, or both, as shown in FIG. 2. In some embodiments, the sensor or sensors **92** are located in the outer casing **19** radially outward of the fan blades **22** and vanes **30**. The at least one sensor **92** may include one of or a combination of dynamic sensors, static wall pressure sensors, altitude sensors, sensors configured to detect the angle of attack of the plurality of fan blades **22**, sensors configured to detect the tip timing of the plurality of fan blades **22**, and airspeed sensors. In some embodiments, the sensor **92** may be a dynamic pressure transducer. The sensor **92** may also be a sensor configured to measure a rotational speed of the fan blades **22**, which could be used along with an additional sensor that is a dynamic pressure transducer.

(73) In the illustrative embodiment, the functionality of the control system **90** described herein may

be implemented in various processing and computing devices, and may be located within the engine **10** or outside of the engine **10**. Moreover, the functionality may be configured to operate on executable software provided on the processing and computing devices.

(74) Furthermore, the functionality disclosed herein may be implemented in various configurations using computer-readable media for carrying or having computer-executable instructions or data structures stored thereon. Such computer-readable media can be any available media that can be accessed by a general purpose or special purpose computer. By way of example, and not limitation, such computer-readable media can comprise physical storage and/or memory media such as RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code means in the form of computer-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer. Computer-executable instructions comprise, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing device to perform a certain function or group of functions.

(75) A person skilled in the art will understand that all features and components of all embodiments described herein, including the guide vane assembly **28** described above and the outlet guide vane assemblies described below can be interchanged and modified to include some elements of some embodiments and other elements of other embodiments. For example, even though some embodiments are shown without a static central portion, these embodiments may include such a central portion, as described herein. Similarly, although some embodiments described two tip segments and two hub segments, some embodiments can include one tip and two hub segments, two tip and one hub segments, each with or without a static central hub section, or other combinations thereof. Similarly, the actuation assemblies can be interchanged, with some embodiments and configurations including pass-through actuation assemblies, two assemblies arranged at the tip and hub, different configurations of segmented and full annular rings, and the like.

(76) Another embodiment of an outlet guide vane assembly **128** is shown in FIGS. **8-12**. The outlet guide vane assembly **128** is similar to the outlet guide vane assembly **28** shown in FIGS. **1-7** and described herein. Accordingly, similar reference numbers in the **100** series indicate features that are common between the outlet guide vane assembly **128** and the outlet guide vane assembly **28**. The description of the outlet guide vane assembly **28** is incorporated by reference to apply to the outlet guide vane assembly **128**, except in instances when it conflicts with the specific description and the drawings of the outlet guide vane assembly **128**.

(77) Similar to the outlet guide vane assembly **28** described above, the outlet guide vane assembly **128** includes first tip and hub segments **132**, **142**. In this embodiment, an air manipulating member **190** may be arranged in the small radial gap between the first tip and hub segments **132**, **142**. In particular, in some embodiments, the air manipulating member **190** may be configured as a winglet **190** or platform coupled to an axially forward side **153** of the fixed aft portion **150**, as shown in FIGS. **9-11**, or may be directly coupled to one of the sides **136**, **145** of the tip and hub segments **132**, **142**. In other embodiments, the air manipulating member **190** may be configured as a seal **190S** coupled to the axially forward side **153** of the fixed aft portion **150**, or may be directly coupled to one of the sides **136**, **145** of the tip and hub segments **132**, **142**.

(78) As shown in FIGS. **9-11**, the air manipulating member **190** may be configured as a winglet **190** that reduce radial flows across the tip and hub segments **132**, **142**. This is particularly helpful when the tip and hub segments **132**, **142** are set to non-neutral, opposing pitch angles. As shown in FIG. **9**, the winglet **190** may be generally planar and include a radially outer winglet surface **191** and a radially inner winglet surface **192**, each surface generally facing radially. The winglet **190** further includes an axially aft end **193**, a suction side edge **194**, and a pressure side edge **195**. The suction and pressure side edges **194**, **195** may be curved similarly to the curvature of the airfoil shape of

the vane **130**.

(79) The winglet **190** further includes a forward edge **196** that is curved. In some embodiments, the forward edge **196** is curved to match the rotational path, or path of movement, of the leading edge **133**, **143** of the tip and hub segments **132**, **142**. In particular, the forward edge **196** may include a radius of curvature **197** that matches the rotational path of the tip and hub segments **132**, **142**, as shown in FIG. **9**. In some embodiments, a width **190W** of the winglet **190** is equal to or greater than a width **30W** of the first tip segment **132** and the first hub segment **142** along an entire axial extent of the first tip segment **132** and the first hub segment **142**. In some embodiments, the width of the winglet **190** at the axially aft end **193** is less than a length of the forward edge **196**. In some embodiments, the winglet **190** has a radial height that allows for the seal to barely contact or nearly contact the sides **136**, **145** of the tip and hub segments **132**, **142** so as to prevent flow from entering this gap.

(80) FIG. **10** and FIG. **11** show a side and front perspective view, respectively, of the winglet **190**. In the example shown in FIG. **10**, the tip segment **132** is moved to a first pitch angle, which is to the left of a neutral zero angle position when viewing FIG. **10** and FIG. **11**. As can be seen, the forward edge **196** is long enough such that the pressure side edge **195** is located beyond the position of the tip segment **132** in the circumferential direction. In some embodiments, the forward edge **196** is formed to be long enough such that both the suction and pressure side edges **194**, **195** are located outside of the potential range of pitch angle positions of the tip and hub segments **132**, **142**. In other words, the path of movement has a maximum first end point and a maximum second end point, and the length of the forward edge **196** is equal to or extends beyond the maximum first end point and the maximum second end point of the path of movement of the first tip segment **132** and/or and the first hub segment **142**. This ensures that radial flows across the tip and hub segments **132**, **142** will continue to be better matched at all pitch angle positions of the tip and hub segments **132**, **142**.

(81) As can be seen in FIG. **12**, the air manipulating member **190** may be configured as a seal **190S** which is coupled to the axially forward side **153** of the fixed aft portion **150** at an axially aft end **193S** of the seal **190S**. The seal **190S** may be generally planar. In some embodiments, the seal **190S** has a radial height that allows for the seal to barely contact or nearly contact the sides **136**, **145** of the tip and hub segments **132**, **142** so as to seal the radial gap between the segments **132**, **142** and prevent flow from entering this gap. In some embodiments, the seal **190S** may be directly coupled to the radially inner side **136** of the tip segment **132**. In some embodiments, the seal **190S** may be directly coupled to the radially outer side **145** of the hub segment **142**. In some embodiments, a width **190SW** of the seal **190S** is equal to or greater than a width **30W** of the first tip segment **132** and the first hub segment **142** along an entire axial extent of the first tip segment **132** and the first hub segment **142**. In some embodiments, the seal **190S** may include a cutout **198** so as to allow a hinge rod **137** to pass therethrough. The winglet **190** may include a similar cutout **198**. In other embodiments in which the segments **132**, **142** do not include a hinge rod, the cutout **198** would not be necessary. In some embodiments, an outer perimeter of the seal **190S** generally corresponds to an outer perimeter of each of the first tip segment **132** and the first hub segment **142**, as shown in FIG. **12**.

(82) Another embodiment of an outlet guide vane assembly **228** is shown in FIG. **13**. The outlet guide vane assembly **228** is similar to the outlet guide vane assemblies **28**, **128** shown in FIGS. **1-12** and described herein. Accordingly, similar reference numbers in the **200** series indicate features that are common between the outlet guide vane assembly **228** and the outlet guide vane assemblies **28**, **128**. The descriptions of the outlet guide vane assemblies **28**, **128** are incorporated by reference to apply to the outlet guide vane assembly **228**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **228**.

(83) Similar to the outlet guide vane assembly **228** described above, the outlet guide vane assembly **228** includes first tip and hub segments **232**, **242**. In this embodiment, the first variable leading

edge outlet guide vane **230** further includes a central portion **248** arranged between the first tip segment **232** and the first hub segment **242** such that the first tip segment **232** and the first hub segment **242** are radially spaced apart. In some embodiments, the central portion may be coupled to and extend axially away from the axially forward side **253** of the fixed aft portion **250**. The central portion **248** is configured to be static and not rotate.

(84) As can be seen in FIG. **13**, the central portion **248** includes a radially outer side **248A**, a radially inner side **248B**, and an axially forward end **248C** extending between the radially outer and inner sides **248A**, **248B** and partially defining the leading edge of the vane **230** along with the tip and hub segments **232**, **242**. The central portion **248** may further include a radially outer hinge rod receiving cavity **249A** configured to receive a hinge rod **237** of tip segment **232**, and a radially inner hinge rod receiving cavity **249B** configured to receive a hinge rod **247** of hub segment **242**. The central portion **248** may provide stability to the tip and hub segments **232**, **242**, as well as allow for uninterrupted flow over the central portion of the vane **230**.

(85) Another embodiment of an outlet guide vane assembly **328** is shown in FIG. **14**. The outlet guide vane assembly **328** is similar to the outlet guide vane assemblies **28**, **128**, **228** shown in FIGS. **1-13** and described herein. Accordingly, similar reference numbers in the **300** series indicate features that are common between the outlet guide vane assembly **328** and the outlet guide vane assemblies **28**, **128**, **128**, **228**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228** are incorporated by reference to apply to the outlet guide vane assembly **328**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **328**.

(86) Similar to the outlet guide vane assembly **228** described above, the outlet guide vane assembly **328** includes first tip and hub segments **332**, **342** and a central portion **348**. Unlike the outlet guide vane assembly **228**, the outlet guide vane assembly **328** may include a single or multiple air manipulating members **390A**, **390B** arranged in the small radial gaps between the tip segment **332** and central portion **348** and between the hub segment **342** and central portion **348**. The air manipulating members **390A**, **390B** may be formed similarly to the air manipulating member **190** described above, in particular formed as a seal or winglet.

(87) In some embodiments, one of the air manipulating members **390A**, **390B** may be formed as a winglet while the other is formed as a seal. In some embodiments, both of the air manipulating members **390A**, **390B** are formed as a seal. In some embodiments, both of the air manipulating members **390A**, **390B** are formed as a winglet. In embodiments in which the tip and hub segments **332**, **342** are rotationally attached to the central portion **348** via hinge rods similar to those described above, the air manipulating members **390A**, **390B** can include a cutout (not shown but similar to the cutout **198** shown in FIG. **11**) to allow the hinge rods to pass therethrough. In other arrangements in which the tip and hub segments **332**, **342** are entirely spaced apart from the central portion **348**, similar to embodiment shown in FIG. **7**, the cutout would not be necessary.

(88) As can be seen in FIG. **15**, the forward edge **396A** of the winglet **390A** is long enough such that the pressure side edge **395A** is located beyond the position of the tip segment **332** in the circumferential direction. Similarly, the forward edge **396B** of the winglet **390B** is long enough such that the suction side edge **394B** is located beyond the position of the hub segment **342** in the circumferential direction. In some embodiments, the forward edges **396A**, **396B** are formed to be long enough such that both the suction and pressure side edges **394A**, **394B**, **395A**, **395B** are located outside of the potential range of pitch angle positions of the tip and hub segments **332**, **342**. This ensures that radial flows across the tip and hub segments **332**, **342** will continue to be better matched at all pitch angle positions of the tip and hub segments **332**, **342**.

(89) Another embodiment of an outlet guide vane assembly **428** is shown in FIGS. **16-19**. The outlet guide vane assembly **428** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328** shown in FIGS. **1-15** and described herein. Accordingly, similar reference numbers in the **400** series indicate features that are common between the outlet guide vane assembly **428** and the outlet guide vane assemblies **28**, **128**, **128**, **228**, **328**. The descriptions of the outlet guide vane assemblies

28, 128, 228, 328 are incorporated by reference to apply to the outlet guide vane assembly **428**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **428**.

(90) Similar to the outlet guide vane assembly **28** described above, the outlet guide vane assembly **428** includes first tip and hub segments **432, 442**. Unlike the vane assemblies **28, 128, 228, 328** described above, the vane assembly **428** only includes a single actuation assembly **470**, in particular a first actuation assembly **470** as shown in FIG. **16A**. The single actuation assembly **470** is configured to independently rotate both the first tip segment **432** and the first hub segment **442**. As such, the first hub segment **442** is rotatably coupled to the inner wall **23** via a hub hinge **23H** arranged within a hinge receptacle **23R** formed in a radially outer side of the inner wall **23**.

(91) As can be seen in FIG. **16A**, the first actuation assembly **470** is arranged radially outward of the leading edge portion **431**, in particular radially outward of the radially outer side **435** of the first tip segment **432**, and includes a first actuation head **476** coupled to the first tip segment **432**, a second actuation head **484**, and a first actuation rod **478** coupled to the second actuation head **484** and to the first hub segment **432**.

(92) Specifically, the first actuation head **476** has a first cavity **479** formed therethrough that is substantially similarly shaped to the outer surface of the first actuation rod **478**. Similarly, the first tip segment **432** includes a second cavity **437** extending therethrough. The second cavity **437** is axially aligned with the first cavity **479**, and in particular, both cavities **437, 479** are axially aligned with the leading edge pitch axis **439**. As a result, as shown in FIG. **16A**, the first actuation rod **478** can extend from the second actuation head **484**, through the first cavity **479**, through the second cavity **437**, and couple to the radially outer side **445** of the first hub segment **442**.

(93) In this configuration, the first actuation head **476** is configured to selectively rotate the first tip segment **432** about the leading edge pitch axis **439** to a first pitch angle relative to the incoming fan exit air **15** utilizing similar actuation methodologies as described above, such as via the actuation arm **474** in conjunction with the annular ring **462**, ring segments, or individual actuators. The second actuation head **484** is configured to selectively rotate the first hub segment **442** via rotation thereof, which rotates the first actuation rod **478** and thus the first hub segment **442**. The first hub segment **442** is rotated independent of the first tip segment **432** about the leading edge pitch axis **439** to a second pitch angle relative to the incoming fan exit air **15**. As such, the tip and hub segments **432, 442** can be rotated to the same or different pitch angles based on the design and operation requirements, such as the operating conditions, predetermined angles, and the like described above.

(94) In some embodiments, the first actuation assembly **470** can further include an actuator coupled to the second actuation head **484** that is configured to rotate the second actuation head **484**. In some embodiments, the actuator may be a relatively small hydraulic actuator or an electric motor actuator such as a stepper motor directly coupled to the second actuation head **484**. In other embodiments, the second action head **484** may be actuated via similar means as to the first actuation head **476**, such as via a further actuation arm coupled to an annular ring, ring segments, or individual actuators.

(95) For example, as shown in the alternative actuation arrangement of the vane assembly **428'** of FIG. **16B**, in some embodiments, the second actuation head **484'** is rotated via pivoting of a similar actuation arm **485'** and annular ring **464'** coupled thereto. The annular ring **464'** may be fully annular or segmented. In such an embodiment, the actuation arm **485'** extends axially forward opposite of the direction of extension of the actuation arm **474'**. The annular ring **464'** is arranged axially forward of the annular ring **462'**.

(96) Similarly, as shown in the alternative actuation arrangement of the vane assembly **428''** of FIG. **16C**, in some embodiments, the second actuation head **484''** is rotated via pivoting of a similar actuation arm **485''** and annular ring **464''** coupled thereto. The annular ring **464''** may be fully annular or segmented. In such an embodiment, the actuation arm **485''** extends axially aft in the

same direction of extension of the actuation arm **474"**. The annular ring **464"** is arranged radially outward of the annular ring **462"**, and may be axially forward of the annular ring **462"**, as shown in FIG. **16C**. In some embodiments, it may be feasible to arrange the annular ring **464"** axially aft of the annular ring **462"**.

(97) As shown in FIG. **17**, the diameter of the first cavity **479** formed in the first actuation head **476** is smaller than the overall diameter of the first actuation head **476**. For reference, FIG. **17** shows a rotation indicator **480** for the first actuation head **476** and a rotation indicator **481** for the first actuation rod **478**. Using these indicators **480**, **481**, the relative rotation of the first actuation head **476** and the first actuation rod **478** can be visualized. For example, FIG. **17** shows the indicators **480**, **481** aligned with each other such that the first tip and hub segments **432**, **442** are at a neutral, zero-angle position. FIG. **18** shows the first hub segment **442** rotated to a first pitch angle while the first tip segment **432** remains at a neutral position, which can be seen by the offset in the indicators **480**, **481**. FIG. **19** shows the first tip segment **432** rotated to a first pitch angle while the first hub segment **442** remains at a neutral position, which can be seen by the offset in the indicators **480**, **481**.

(98) Another embodiment of an outlet guide vane assembly **528** is shown in FIG. **20**. The outlet guide vane assembly **528** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428** shown in FIGS. **1-19** and described herein. Accordingly, similar reference numbers in the **500** series indicate features that are common between the outlet guide vane assembly **528** and the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428** are incorporated by reference to apply to the outlet guide vane assembly **528**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **528**.

(99) Similar to the outlet guide vane assembly **428** described above, the outlet guide vane assembly **528** includes first tip and hub segments **532**, **542** and a single actuation assembly **570** configured to move both tip and hub segments **532**, **542**. In some embodiments, the first actuation assembly **570** can include the first actuation arm **574**, as well as an actuator coupled to the second actuation head **584** that is configured to rotate the second actuation head **584**. In some embodiments, the actuator may be a relatively small hydraulic actuator or an electric motor actuator such as a stepper motor directly coupled to the second actuation head **584**. In other embodiments, the second actuation head **584** may be actuated via similar means as to the first actuation head **576**, and similar means as the section actuation head **484**, such as via a further actuation arm coupled to an annular ring, ring segments, or individual actuators, such as those shown in FIG. **16B** and FIG. **16C**.

(100) In this embodiment, an air manipulating member **590** similar to the air manipulating members **190**, **390A**, **390B** described above may be arranged in the small radial gap between the first tip and hub segments **532**, **542**. In particular, in some embodiments, the air manipulating member **590** may be configured as a winglet **590** or platform coupled to an axially forward side **553** of the fixed aft portion **550**. In other embodiments, the air manipulating member **590** may be configured as a seal **590S**.

(101) Similar to the air manipulating member **190**, the air manipulating member **590** may be configured as a winglet **590** that reduce radial flows across the tip and hub segments **532**, **542**. Also similar to the air manipulating member **190**, the air manipulating member **590** may be configured as a seal **590S** which is coupled to the axially forward side **553** of the fixed aft portion **550** or directly coupled to one of the radially inner side **536** of the tip segment **532** or the radially outer side **545** of the hub segment **542**.

(102) Another embodiment of an outlet guide vane assembly **628** is shown in FIG. **21**. The outlet guide vane assembly **628** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528** shown in FIGS. **1-20** and described herein. Accordingly, similar reference numbers in the **600** series indicate features that are common between the outlet guide vane assembly **628** and the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**. The descriptions of the outlet guide vane

assemblies **28**, **128**, **228**, **328**, **428**, **528** are incorporated by reference to apply to the outlet guide vane assembly **628**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **628**.

(103) Similar to the outlet guide vane assembly **428** described above, the outlet guide vane assembly **628** includes first tip and hub segments **632**, **642** and a single actuation assembly **670** configured to move both tip and hub segments **632**, **642**. In some embodiments, the first actuation assembly **670** can include the first actuation arm **674**, as well as an actuator coupled to the second actuation head **684** that is configured to rotate the second actuation head **684**. In some embodiments, the actuator may be a relatively small hydraulic actuator or an electric motor actuator such as a stepper motor directly coupled to the second actuation head **684**. In other embodiments, the second actuation head **684** may be actuated via similar means as to the first actuation head **676**, and similar means as the section actuation head **484**, such as via a further actuation arm coupled to an annular ring, ring segments, or individual actuators, such as those shown in FIG. **16B** and FIG. **16C**.

(104) In this embodiment, the first variable leading edge outlet guide vane **630** further includes a central portion **648** arranged between the first tip segment **632** and the first hub segment **642** such that the first tip segment **632** and the first hub segment **642** are radially spaced apart. In some embodiments, the central portion **648** may be coupled to and extend axially away from the axially forward side **653** of the fixed aft portion **650**.

(105) As can be seen in FIG. **21**, the central portion **648** includes a radially outer side **648A**, a radially inner side **648B**, and an axially forward end **648C** extending between the radially outer and inner sides **648A**, **648B** and partially defining the leading edge of the vane **630** along with the tip and hub segments **632**, **642**. The central portion **648** may further include a radially extending third cavity **648D** that is axially aligned with the second cavity **637** and the first cavity **679** formed in the first actuation head **676**. As such, the first actuation rod **678** can extend from the second actuation head **684**, through the first actuation head **676**, through the first tip segment **632**, through the central portion **648**, and couple to the radially outer side **645** of the hub segment **642** such that the second actuation head **684** can rotate the hub segment **642**. The central portion **648** may provide stability to the tip and hub segments **632**, **642**, as well as allow for uninterrupted flow over the central portion of the vane **630**.

(106) Another embodiment of an outlet guide vane assembly **728** is shown in FIG. **22**. The outlet guide vane assembly **728** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628** shown in FIGS. **1-21** and described herein. Accordingly, similar reference numbers in the **700** series indicate features that are common between the outlet guide vane assembly **728** and the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628** are incorporated by reference to apply to the outlet guide vane assembly **728**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **728**.

(107) Similar to the outlet guide vane assemblies **528**, **628** described above, the outlet guide vane assembly **728** includes first tip and hub segments **732A**, **742A** and a central portion **748**. Unlike the assemblies described above, the outlet guide vane assembly **728** further includes a second tip segment **732B** and a second hub segment **742B** arranged radially between the first tip segment **732A** and the central portion **748** and between the first hub segment **742A** and the central portion **748**, respectively. The additional second tip segment **732B** and second hub segment **742B** allow for additional variations of the fan exit air **15** flowing over the vane **730**.

(108) In order to move all four segments **732A**, **732B**, **742A**, **742B**, the assembly **728** includes a first actuation assembly **770A** and a second actuation assembly **770B** that may be configured substantially similarly to each other. For example, the first actuation head **776A** of the first actuation assembly **770A** has a first cavity **779A** formed therethrough. Similarly, the first tip segment **732A** includes a second cavity **737A** extending therethrough. The second cavity **737A** is

axially aligned with the first cavity **779A**. As a result, as shown in FIG. 22, the first actuation rod **778A** of the first actuation assembly **770A** can extend from the second actuation head **784A**, through the first cavity **779A**, through the second cavity **737A**, and couple to the radially outer side **735B** of the second tip segment **732B**. As such, the second actuation head **784A** can rotate the second tip segment **732B** independent of the first tip segment **732A**, which is rotated by the first actuation head **776A**.

(109) Similarly, the first actuation head **776B** of the second actuation assembly **770B** has a first cavity **779B** formed therethrough. Similarly, the first hub segment **742A** includes a second cavity **747A** extending therethrough. The second cavity **747A** is axially aligned with the first cavity **779B**. As a result, as shown in FIG. 22, the first actuation rod **778B** of the second actuation assembly **770B** can extend from the second actuation head **784B**, through the first cavity **779B**, through the second cavity **747A**, and couple to the radially inner side **746B** of the second hub segment **742B**. As such, the second actuation head **784B** can rotate the second hub segment **742B** independent of the first hub segment **742A**, which is rotated by the first actuation head **776B**. The central portion **748** can further include hinge rods **748D**, **748E** extending away from its radially outer and inner sides **748A**, **748B**, respectively, which are received in hinge rod receiving cavities **737B**, **747B** of the second tip and hub segments **732B**, **742B**, respectively.

(110) Each actuation rod **778A**, **778B** has a central axis. In some embodiments, due to the curvature of the leading edge of the leading edge portion **31**, the central axes of the actuation rods **778A**, **778B** may need to be offset to accommodate the curvature and the offset leading edge pitch axes **739A**, **739B** of the various segments. For example, as shown in FIG. 22, the first hub segment **742A** may have a leading edge pitch axis **739B** offset from the leading edge pitch axis **739A** of the three radially outer segments **732A**, **732B**, **742B**. As such, the actuation rod **778B** of the second actuation assembly **770B** must be aligned with the leading edge pitch axis **739B** so as to effectively rotate the first hub segment **742A**. In other embodiments, however, the curvature of the leading edge may not have as great of an effect, and both actuation rods **778A**, **778B** can be aligned with a singular leading edge pitch axis **739A** that all four segments **732A**, **732B**, **742A**, **742B** rotate about.

(111) Another embodiment of an outlet guide vane assembly **828** is shown in FIG. 23 and FIG. 24. The outlet guide vane assembly **828** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628**, **728** shown in FIGS. 1-22 and described herein. Accordingly, similar reference numbers in the **800** series indicate features that are common between the outlet guide vane assembly **828** and the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628**, **728**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628**, **728** are incorporated by reference to apply to the outlet guide vane assembly **828**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **828**.

(112) Similar to the outlet guide vane assembly **728** described above, the outlet guide vane assembly **828** includes first and second tip and hub segments **832A**, **832B**, **842A**, **842B**, a central portion **848**, and two actuation assemblies **870A**, **870B** configured to move the first and second tip and hub segments **832A**, **832B**, **842A**, **842B** independently of each other. In this embodiment, an air manipulating member **890** similar to the air manipulating members **190**, **390A**, **390B**, **590** described above may be arranged in the small radial gaps between the first and second tip and hub segments **832A**, **832B**, **842A**, **842B** and between the second tip and hub segments **832B**, **842B** and the central portion **848**. In particular, in some embodiments, the air manipulating members **890** may be configured as a winglets **890** or platforms coupled to an axially forward side **853** of the fixed aft portion **850**. In other embodiments, the air manipulating members **890** may be configured as seals **890S**.

(113) Similar to the air manipulating members **190**, **390A**, **390B**, **590**, the air manipulating members **890** may be configured as winglet **890** that reduce radial flows across the first and second tip and hub segments **832A**, **832B**, **842A**, **842B**, as shown in FIG. 24. Also similar to the air manipulating members **190**, **390A**, **390B**, **590**, the air manipulating member **890** may be configured

as seals **890S** coupled to the axially forward side **853** of the fixed aft portion **850** or directly coupled to one of the sides **836A**, **836B**, **845A**, **845B** of the first and second tip and hub segments **832A**, **832B**, **842A**, **842B**.

(114) Another embodiment of an outlet guide vane assembly **928** is shown in FIG. 25. The outlet guide vane assembly **928** is similar to the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628**, **728**, **828** shown in FIGS. 1-24 and described herein. Accordingly, similar reference numbers in the **900** series indicate features that are common between the outlet guide vane assembly **928** and the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628**, **728**, **828**. The descriptions of the outlet guide vane assemblies **28**, **128**, **228**, **328**, **428**, **528**, **628**, **728**, **828** are incorporated by reference to apply to the outlet guide vane assembly **928**, except in instances when they conflict with the specific description and the drawings of the outlet guide vane assembly **928**.

(115) Similar to the outlet guide vane assembly **728** described above, the outlet guide vane assembly **928** includes first and second tip and hub segments **932A**, **932B**, **942A**, **942B** and two actuation assemblies **970A**, **970B** configured to move the first and second tip and hub segments **932A**, **932B**, **942A**, **942B** independently of each other. Unlike the outlet guide vane assembly **728**, the vane assembly **928** does not include a central portion **948**. As such, the second hub segment **942B** can include a hinge rod **947** extending radially away from the segment **942B** and received in a hinge rod receiving cavity **937B** of the second tip segment **932B**.

(116) It should be appreciated that features and elements of all of the embodiments described above can be used in conjunction with alternative embodiments that would be understood by one of ordinary skill in the art. For example, a vane having three rotatable segments could be configured that includes two radially outer segments that are movable via an actuation assembly similar to the single actuation assembly **470** described above and a single radially inner segment movable via a second actuation assembly. Or, more than four segments may be utilized in a vane arrangement. Such similar alternative embodiments are contemplated by the present disclosure.

(117) FIGS. 26-30 are graphs showing nominal and variation profiles of a particular outlet guide vane including four moveable segments and a static central portion, such as the embodiments shown in FIGS. 22-24. The graphs show span percentage versus inlet vane angle in degrees, with negative degrees indicating the vane profile is more open and positive degrees indicating the vane profile is more closed. The profiles shown in FIG. 26 are of a nominal vane versus a generalized matched vane. FIGS. 27, 28, and 30 are related to a particular outlet guide vane including four moveable segments and a static central portion, while the profiles shown in FIG. 29 are related to a particular outlet guide vane including two moveable segments and a static central portion.

(118) The nominal vane cannot be varied, but each of the segmented vane's segments may be rotated so that the distribution of the segmented vane resembles the variation profile. FIG. 27 displays the variant distribution versus rotated segmented portions of the nominal profile to resemble the variant distribution. FIG. 28 displays the rotated segments of FIG. 27 but removes the variant distribution of FIG. 27. It is noted that the nominal and variant distribution are nearest to each other at approximately mid-span, around 50% to 70%. This may be true of other variant distributions where they are near the nominal or could cross so therefore it is logical that the static portion of the vane be at such a spanwise location. Additionally, the largest incidence delta from nominal may occur at the hub or tip which results in the most benefit for adjusting the leading edge angle would be here instead of midspan. The shift to around 60% rather than purely midspan for the center of the static portion also makes sense as the area is greater at higher spans compared to inner spans, this would be a beneficial approach for maintaining flow. A balance between flow requirements and incidence reduction must be considered for the segmented vane versus the nominal vane.

(119) Moreover, as can be seen in FIGS. 26-28, the variations may be within 5 degrees opened or closed (i.e. not a large setting change), which is meaningful in terms of stall or performance. It is also possible to achieve a segmented vane to be within 3 degrees. These values would make it

possible to have the winglets described above at the inner ends of segments to help avoid discontinuities between segments. FIG. 29 shows a two-segment vane having a static central portion, where the vane segments are rotated to closely adhere to the variation profile. FIG. 30 shows another four segment and static central portion vane over the span, with variations possible within 5 degrees open (negative) or closed (positive). The control system 90 can be configured to maintain overall choke margin even with different circumferential extents varying tip and hub segments by various degrees to match their setting angles to inlet flows.

(120) A method can include arranging a fan duct 20 circumferentially around a central axis 11, providing a fan 21 comprising a plurality of fan blades 22 that extend radially outward relative to the central axis 11 and that are adapted to rotate about the central axis 11 to force fan exit air 15 toward an aft end of the fan duct 20, and arranging an outlet guide vane assembly 128 in the fan duct 20 axially downstream of the fan 21 and configured to adjust a direction of incoming fan exit air 15 received from the plurality of fan blades 22 and reduce incidence between the outlet guide vane assembly 128 and the fan exit air 15, the outlet guide vane assembly 128 including a first variable leading edge outlet guide vane 130 extending radially relative to the central axis 11 and including the leading edge portion 131 and a fixed aft portion 150, the leading edge portion 131 including a first tip segment 132 configured to rotate about a leading edge pitch axis 139 and a first hub segment 142 located radially inward of and separate from the first tip segment 132, the first hub segment 142 configured to independently rotate about the leading edge pitch axis 139 relative to the first tip segment 132.

(121) The method can further include arranging a first air manipulating member 190 radially between the first tip segment 132 and the first hub segment 142 and configured to manipulate at least a portion of the incoming fan exit air 15 flowing over the leading edge portion 131, and arranging at least one actuation assembly 170 relative to the first tip segment 132 configured to selectively rotate the first tip segment 132 about the leading edge pitch axis 139 to a first pitch angle relative to the incoming fan exit air 15 and configured to selectively rotate the first hub segment 142 independent of the first tip segment 132 about the leading edge pitch axis 139 to a second pitch angle relative to the incoming fan exit air 15.

(122) While the disclosure has been illustrated and described in detail in the foregoing drawings and description, the same is to be considered as exemplary and not restrictive in character, it being understood that only illustrative embodiments thereof have been shown and described and that all changes and modifications that come within the spirit of the disclosure are desired to be protected. Moreover, each of the embodiments described herein, including the specific features and characteristics of each embodiment, may be combined with other embodiments as would be understood by one of ordinary skill in the art.

Claims

1. A fan assembly for a gas turbine engine the fan assembly comprising a fan duct arranged circumferentially around a central axis, a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades and reduce incidence between the outlet guide vane assembly and the fan exit air, the outlet guide vane assembly including a first variable leading edge outlet guide vane that extends radially relative to the central axis and includes a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, a first air manipulating member

arranged radially between the first tip segment and the first hub segment and configured to manipulate at least a portion of the incoming fan exit air flowing over the leading edge portion, and at least one actuation assembly configured to selectively rotate the first tip segment about the leading edge pitch axis to a first pitch angle relative to the incoming fan exit air and configured to selectively rotate the first hub segment independent of the first tip segment about the leading edge pitch axis to a second pitch angle relative to the incoming fan exit air, wherein the first air manipulating member is arranged proximate to a radially inner side of the first tip segment and proximate to a radially outer side of the first hub segment, wherein a radial gap is formed between the radially inner side of the first tip segment and the radially outer side of the first hub segment, wherein the first air manipulating member has a radial extent that is sized so as to seal off the incoming fan exit air from flowing through the radial gap, wherein the first air manipulating member is a winglet defining a radially outer surface that is planar and faces radially outwardly and a radially inner surface that is planar and faces radially inwardly, wherein the winglet includes an axially aft end and an axially forward edge, wherein a width of the winglet is equal to or greater than a width of the first tip segment and the first hub segment along an entire axial extent of the first tip segment and the first hub segment, wherein a width of the winglet at the axially aft end is less than a length of the forward edge, wherein the forward edge is curved so as to match a path of movement of a leading edge of at least one of the first tip segment and the first hub segment, wherein the path of movement has a maximum first end point and a maximum second end point, and wherein the length of the forward edge is equal to or extends beyond the maximum first end point and the maximum second end point of the path of movement of the at least one of the first tip segment and the first hub segment.

2. The fan assembly of claim 1, wherein the at least one actuation assembly is configured to rotate the first tip segment and the first hub segment to the first pitch angle and the second pitch angle which is different than the first pitch angle.

3. The fan assembly of claim 2, wherein the at least one actuation assembly includes a first actuation assembly arranged radially outward of the leading edge portion and configured to rotate the first tip segment to the first pitch angle.

4. The fan assembly of claim 3, wherein the at least one actuation assembly further includes a second actuation assembly arranged radially inward of the leading edge portion and configured to rotate the first hub segment to the second pitch angle.

5. The fan assembly of claim 4, wherein the at least one actuation assembly includes a first actuation head coupled to the first tip segment and a first actuation arm coupled to the first actuation head, wherein rotation of the first actuation arm causes rotation of the first actuation head which causes rotation of the first tip segment so as to rotate the first tip segment to the first pitch angle, wherein the second actuation assembly includes a second actuation head coupled to the first hub segment and a second actuation arm coupled to the second actuation head, and wherein rotation of the second actuation arm causes rotation of the second actuation head which causes rotation of first hub segment so as to rotate the first hub segment independent of the first tip segment to the second pitch angle.

6. The fan assembly of claim 1, wherein the aft end of the first air manipulating member is coupled to an axially forward end of the fixed aft portion.

7. A fan assembly for a gas turbine engine the fan assembly comprising a fan duct arranged circumferentially around a central axis, a fan adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, and an outlet guide vane assembly located in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades and reduce incidence between the outlet guide vane assembly and the fan exit air, the outlet guide vane assembly including a first variable leading edge outlet guide vane including a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment

located radially inward of and separate from the first tip segment, a first air manipulating member arranged radially between the first tip segment and the first hub segment, the first air manipulating member including a forward end proximate to a leading edge of the leading edge portion of the first variable leading edge outlet guide vane and an aft end opposite the forward end, the forward end having a width measured in a circumferential direction that is greater than a width of the aft end so as to form a winglet shape, and at least one actuation assembly configured to selectively rotate the first tip segment about the leading edge pitch axis to a first pitch angle relative to the incoming fan exit air and configured to selectively rotate the first hub segment independent of the first tip segment about the leading edge pitch axis to a second pitch angle relative to the incoming fan exit air, wherein the first air manipulating member is arranged proximate to a radially inner side of the first tip segment and proximate to a radially outer side of the first hub segment, wherein a radial gap is formed between the radially inner side of the first tip segment and the radially outer side of the first hub segment, and wherein the first air manipulating member has a radial extent that is sized so as to seal off the incoming fan exit air from flowing through the radial gap, wherein the first air manipulating member defines a radially outer surface that is planar and faces radially outwardly and a radially inner surface that is planar and faces radially inwardly, and wherein a width of the first air manipulating member is equal to or greater than a width of the first tip segment and the first hub segment along an entire axial extent of the first tip segment and the first hub segment, and wherein the forward end includes a forward edge that is curved so as to match a path of movement of a leading edge of at least one of the first tip segment and the first hub segment, wherein the path of movement has a maximum first end point and a maximum second end point, and wherein the length of the forward edge is equal to or extends beyond the maximum first end point and the maximum second end point of the path of movement of the at least one of the first tip segment and the first hub segment.

8. The fan assembly of claim 7, wherein the at least one actuation assembly includes a first actuation assembly arranged radially outward of the leading edge portion and configured to rotate the first tip segment to the first pitch angle.

9. A method comprising arranging a fan duct circumferentially around a central axis, providing a fan comprising a plurality of fan blades that extend radially outward relative to the central axis and that are adapted to rotate about the central axis to force fan exit air toward an aft end of the fan duct, arranging an outlet guide vane assembly in the fan duct axially downstream of the fan and configured to adjust a direction of incoming fan exit air received from the plurality of fan blades and reduce incidence between the outlet guide vane assembly and the fan exit air, the outlet guide vane assembly including a first variable leading edge outlet guide vane extending radially relative to the central axis and including a leading edge portion and a fixed aft portion, the leading edge portion including a first tip segment configured to rotate about a leading edge pitch axis and a first hub segment located radially inward of and separate from the first tip segment, the first hub segment configured to independently rotate about the leading edge pitch axis relative to the first tip segment, arranging a first air manipulating member radially between the first tip segment and the first hub segment and configured to manipulate at least a portion of the incoming fan exit air flowing over the leading edge portion, the first air manipulating member including a radially outer surface that is planar from a forward end of the first air manipulating member to an aft end of the first air manipulating member opposite the forward end and that faces radially outwardly, and a radially inner surface that is planar from the forward end to the aft end and that faces radially inwardly, and arranging at least one actuation assembly relative to the first tip segment configured to selectively rotate the first tip segment about the leading edge pitch axis to a first pitch angle relative to the incoming fan exit air and configured to selectively rotate the first hub segment independent of the first tip segment about the leading edge pitch axis to a second pitch angle relative to the incoming fan exit air, wherein the first air manipulating member is arranged proximate to a radially inner side of the first tip segment and proximate to a radially outer side of

the first hub segment, wherein a radial gap is formed between the radially inner side of the first tip segment and the radially outer side of the first hub segment, and wherein the first air manipulating member has a radial extent that is sized so as to seal off the incoming fan exit air from flowing through the radial gap, wherein the first air manipulating member is a winglet defining a radially outer surface that is planar and faces radially outwardly and a radially inner surface that is planar and faces radially inwardly, and wherein the winglet includes an axially aft end and an axially forward edge, wherein a width of the winglet is equal to or greater than a width of the first tip segment and the first hub segment along an entire axial extent of the first tip segment and the first hub segment, and wherein a width of the winglet at the axially aft end is less than a length of the forward edge, wherein the forward edge is curved so as to match a path of movement of a leading edge of at least one of the first tip segment and the first hub segment, wherein the path of movement has a maximum first end point and a maximum second end point, and wherein the length of the forward edge is equal to or extends beyond the maximum first end point and the maximum second end point of the path of movement of the at least one of the first tip segment and the first hub segment.

10. The fan assembly of claim 7, wherein the at least one actuation assembly is configured to rotate the first tip segment and the first hub segment to the first pitch angle and the second pitch angle which is different than the first pitch angle.

11. The method of claim 9, wherein the at least one actuation assembly further includes a second actuation assembly arranged radially inward of the leading edge portion and configured to rotate the first hub segment to the second pitch angle.

12. The method of claim 9, wherein the at least one actuation assembly is configured to rotate the first tip segment and the first hub segment to the first pitch angle and the second pitch angle which is different than the first pitch angle.

13. The fan assembly of claim 7, wherein the aft end of the first air manipulating member is coupled to an axially forward end of the fixed aft portion.

14. The method of claim 9, wherein the aft end of the first air manipulating member is coupled to an axially forward end of the fixed aft portion.

15. The method of claim 9, wherein the at least one actuation assembly includes a first actuation assembly arranged radially outward of the leading edge portion and configured to rotate the first tip segment to the first pitch angle.

16. The fan assembly of claim 7, wherein the at least one actuation assembly further includes a second actuation assembly arranged radially inward of the leading edge portion and configured to rotate the first hub segment to the second pitch angle.
